

# A Parallel Supervision System for Vehicle CO<sub>2</sub> Emissions Based on OBD-Independent Information

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**Abstract**—A parallel supervision system is built in this paper in order to accurately estimate vehicle CO<sub>2</sub> emissions. Only on-board diagnostics (OBD)-independent information is used, making the model capable of making predictions based on future road gradients and planned speed trajectories. Based on the parallel theory, the actual traffic environment is considered the physical world, while the combined CO<sub>2</sub> model (which consists of physical and data-driven models) is the core part of the artificial world. The physical model uses a cascaded structure with engine speeds and torques as intermediate variables, and the data-driven model relies on a modified long short-term memory (LSTM) neural network. When the historical data is sufficient in size and diversity, the data-driven model is appropriate and achieves more accurate estimations; otherwise, the physical model is preferable because of its greater robustness. Based on this combined model, the supervision system can leverage both the learning ability and physics-based knowledge. Two real-world experimental case studies have been performed to validate this system. According to the research analysis, both the physical and data-driven models achieve sufficient accuracy. The physical model indicates more robustness even when some primary parameters (gear ratios) are unknown, which can be used as a supplement to the data-driven model. Moreover, the deterioration factor (DF) of vehicle CO<sub>2</sub> emissions is considered to

simulate aged vehicles. This parallel supervision system can effectively address the gap between regulatory test cycles and real-world carbon emissions.

**Index Terms**—Combined CO<sub>2</sub> estimation model, deterioration factor, OBD-independent, parallel supervision system, vehicle CO<sub>2</sub> emissions.

## I. INTRODUCTION

GREENHOUSE gases (mainly carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), and nitrous oxide (N<sub>2</sub>O)) play a significant role in climate change and affect human lives [1]. Among them, CO<sub>2</sub> contributes the most, accounting for 65% of global emissions [2]. While the transportation sector drives the economy, it is a major source of CO<sub>2</sub> emissions [3]. In China, approximately 10% of CO<sub>2</sub> comes from the transportation sector, totaling approximately 1000 million tons (Mt); this value exceeds the emissions of Germany (659.1 Mt) and the U.K. (339.2 Mt) in 2020 combined [4]. Furthermore, an increase in the vehicle population is inevitable based on China's rapid economic development. Therefore, reducing CO<sub>2</sub> emissions is crucial to achieving China's target of carbon neutrality [5]. Specifically, credible supervision, accurate estimation, and effective optimization are necessary means for reducing carbon emissions from actual traffic flows.

Establishing a supervision system for vehicle CO<sub>2</sub> emissions is beneficial for estimating real-time carbon emissions and formulating effective mitigation strategies [6] and [7]. The core part of the whole system is an accurate estimation method for vehicle CO<sub>2</sub> emissions, which has been investigated by many researchers in recent years. A portable emission measurement system (PEMS) is capable of monitoring and recording real-time vehicle emissions. Such a system is carried in the test vehicle or mounted on a trailer moving along with the test vehicle [8]. However, a PEMS needs some expensive devices and exhibits poor repeatability, limiting its large-scale applicability [9]. Some well-known simulation tools for determining CO<sub>2</sub> emissions and fuel consumption rates (FCRs), such as Virginia Tech's fuel consumption model and MOVES, are calibrated based on the results obtained from standard driving cycles conducted in labs or inspection/maintenance (I/M) procedures [10] and [11]. Although the standard test procedures (mostly conducted in laboratories with chassis dynamometers) are becoming much

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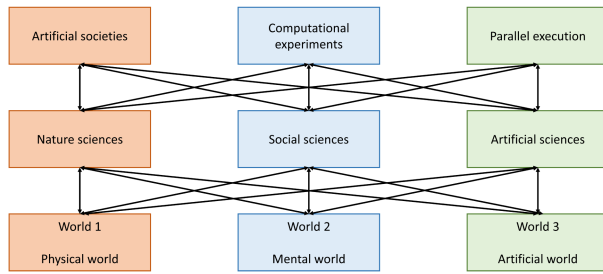


Fig. 1. Conceptual diagram of the ACP approach. [15].

stricter, they are significantly different from real driving conditions [12]. Therefore, this kind of method cannot fully reflect the dynamic characteristics of actual driving scenarios. Moreover, some models use on-board diagnostic (OBD) information as input variables to estimate vehicle CO<sub>2</sub> emissions [13] and [14]. Vehicle-level information (speed, acceleration, road gradient, etc.) and internal engine-level variables (engine speed, torque, throttle position, coolant temperature, etc.) are both available from the OBD interface, which results in a substantial increase in model accuracy. However, specialized equipment is needed to transmit OBD data to a cloud platform, and this becomes costly at a large scale (especially for older vehicles that are mostly not equipped with telematics devices). More importantly, this approach often relies on a high number of internal vehicle variables, which makes it hard to predict future performance (for example, comparing the carbon emissions of vehicles under different predicted speed trajectories). To supervise vehicle CO<sub>2</sub> emissions in actual traffic, developing an accurate estimation model based on vehicle-level predictive information (ahead road conditions and planned speed trajectories) is very meaningful.

The parallel theory aims to model, analyze, and control complex systems, so it can be used as the basis for supervising vehicle carbon emissions [15]. In line with Karl Popper's theory of reality, a parallel system consists of three parts: the physical world, the mental world, and the artificial world. (Fig. 1). Through interactions within the whole system, many complicated tasks can be handled based on the parallel theory [16], [17], and [18]. The physical, mental, and artificial worlds represent real-world scenarios, cognitive behaviors, and digital virtual models, respectively. Artificial societies, computational experiments, and parallel execution (ACP) are the main elements that reveal the operating logic of a parallel system. When applied to the vehicle CO<sub>2</sub> emission supervision, the parallel theory is modified to make it more suitable. The physical world is the actual traffic environment with some necessary sensors, while the artificial world contains a CO<sub>2</sub> emission model to estimate second-by-second emission rates. As highly dynamic processes always lead to high fuel consumption and CO<sub>2</sub> emission levels, high-resolution estimations are beneficial for further spatial-temporal investigations. If the available hardware resources are limited, the estimated frequency may be somewhat reduced. To connect both worlds, reality-virtual interaction is critical, and it has two main functions [19]. On the one hand, it can continuously update the model by comparing the total estimation results with the measurements obtained from roadside intelligent CO<sub>2</sub> sensors.

On the other hand, it is capable of transmitting optimized path plans or driving advice to drivers in real time.

This paper proposes the framework of a parallel supervision system for real-time vehicle CO<sub>2</sub> emissions and develops a combined CO<sub>2</sub> estimation model (containing both physical and data-driven modules). The framework of the parallel supervision system consists of three modules: the physical world, the artificial world, and a cloud communication platform (realizing reality-virtual interaction). This supervision system is capable of estimating the real-time CO<sub>2</sub> emissions from a traffic flow, recognizing high-emission driving patterns, and providing rational advice. Since the associated research is still in progress, this paper mainly focuses on how to develop an accurate and scalable vehicle CO<sub>2</sub> emission model. The accurate estimation of real-time vehicle CO<sub>2</sub> emissions will facilitate spatial-temporal analyses and promote emission-aware eco-driving systems [20]. To adapt to modern traffic scenarios with various powertrains (such as gasoline engines, diesel engines, hybrid vehicles, and electric vehicles), this paper proposes a novel combined vehicle CO<sub>2</sub> emission model. A data-driven model always provides accurate results for the range of observations and has unknown extrapolation properties, so it is more suitable to be adopted as a vehicle-specific model. The physical model exhibits better generalization ability, so it can be used as the category-specific model with acceptable accuracy. When the historical driving data of the vehicles are sufficient and meet the requirements of the training set, the data-driven model is more appropriate. Otherwise, the physical model is preferable. Since the proposed supervision system in this paper is oriented to the entire traffic flow, only simple kinematic variables (e.g., speed and acceleration) and road gradients are used as the model inputs, which also significantly increases the feasibility of subsequent research (such as parallel predictions for optimizing future speed trajectories). In addition, these OBD-independent models can be incorporated into traffic simulation software at different scales [21].

Section II presents a literature review of the existing methods that have been applied to estimate vehicle CO<sub>2</sub> emissions. Section III introduces the technical framework of the proposed parallel supervision system. Section IV outlines the details of both the physical and data-driven CO<sub>2</sub> estimation models. Section V compares the estimations to the real-world experimental findings, and the deterioration factor is also investigated in this section. Finally, the contributions of the study are highlighted in Section VI, along with limitations and future work.

## II. RELATED WORK

Many studies have focused on the estimation of vehicle CO<sub>2</sub> emissions. According to their modeling configurations, the developed methods can be classified into white-box, gray-box, and black-box models based on their transparency [21]. White-box models are mostly based on first principles, which are highly deterministic and accurate [22] and [23]. However, they require a detailed understanding of the given vehicle, even the engine components, which leads to a relatively complex modeling process. In addition, a complete physical model is always too

complex and requires a large number of inputs, which are hard to be obtained in real traffic.

Gray-box models are built using both system knowledge and experimental data, while they can both utilize prior expert knowledge and mine the underlying relationships in historical data. As the formation of CO<sub>2</sub> involves complex combustion, physicochemical, heat and mass transport processes, it exhibits obvious nonlinearity that may be simplified by using a data-driven strategy. Some researchers have adopted a linear hypothesis between wheel power and CO<sub>2</sub> emissions and selected wheel power as a proxy variable. The Veline is a vehicle-specific linear model of this type, which was obtained by fitting the average CO<sub>2</sub> emissions and the average wheel power for the four phases (low, middle, high, and extra-high) within the worldwide harmonized light vehicle test cycle [24]. The authors in [25] further investigated this method, and they offered a unique linear efficiency equation for each gear to achieve improved model accuracy.

Multivariate regression and machine learning (ML) are popular black-box modeling approaches for vehicle CO<sub>2</sub> estimation, while they can estimate system outputs even without any insights about the system. Multivariate regression is suitable for estimating vehicle CO<sub>2</sub> emissions, and some researchers have used engine power or vehicle-specific power (VSP) as the proxy variable [26] and [27]. In [28], the authors adopted regression analysis to predict CO<sub>2</sub> emissions, and they concluded linear relationships between CO<sub>2</sub>, vehicle speed, and acceleration/deceleration. Among these variables, vehicle speed exhibited a stronger correlation with respect to CO<sub>2</sub> emissions. With the development of artificial intelligence technology, ML techniques are taking the lead in data-driven models as they can efficiently capture the complex relationships between multiple inputs and the target output [29] and [30]. However, black-box models always require large amounts of data for training, and they are only valid for the given range of observations with unknown extrapolation properties. The authors in [6] used a long short-term memory (LSTM) model to estimate real-time vehicle CO<sub>2</sub> emissions based on OBD data. As LSTM is a recurrent neural network (RNN) variant that remembers previous information along with the current inputs, this method is suitable for real-time vehicle CO<sub>2</sub> emission estimation.

According to the literature review, ML has been widely used to estimate CO<sub>2</sub> emissions with various network configurations. However, the amounts and diversity levels of the accessible data for various types of vehicles are different. Therefore, if a completely data-driven model is used throughout the whole transportation environment, it will encounter some unfamiliar vehicles, and its extrapolation properties will be challenged. To achieve a trustworthy supervision, such issues should be mitigated by other methods. In addition, the use of OBD data as model inputs has some critical drawbacks. Accessing the OBD interface requires the installation of specialized equipment, raising the cost of large-scale deployment. Third-party spoofing of real-time model inputs is also possible. Moreover, the internal variables derived from OBD are hard to predict, leading to difficult real-time optimization.

To solve these issues, this paper adopts OBD-independent information as the model input variables, significantly

improving the effectiveness and practicability of the whole supervision system. Contrary to pure ML models, a combined model is developed to compensate for insufficient data scenarios. In an actual application, the parallel supervision system selects the appropriate module (the physical module or data-driven module) depending on the availability of vehicle-specific historical data.

### III. PARALLEL SUPERVISION SYSTEM

#### A. Overall Framework

The technical framework of the proposed parallel supervision system is shown in Fig. 2. The whole system consists of three submodules, which are the physical world, the artificial world, and a cloud communication platform. The physical world is the actual traffic environment with various vehicles running on the road, and each vehicle is an individual source of CO<sub>2</sub> emissions. To obtain the real-time model inputs, some roadside sensors should be added to measure speeds, accelerations, and road gradients (in the current experiment, these values are measured by GPS and HD maps). The required data can be divided into three types: simple kinematic variables (speed and acceleration), road conditions (road gradients), and basic vehicle parameters (vehicle mass, engine size, gear box ratios, and mileage). The first two types are mainly acquired from real-time measurements, and the last type is mainly derived from the vehicle certification manual. As vehicle mass is relatively constant, it is periodically updated at highway toll stations.

The combined CO<sub>2</sub> model (including a physical model and a data-driven model) is the core module of the artificial world. By leveraging both the model learning ability and physics-based knowledge, this combined model is capable of estimating real-time vehicle CO<sub>2</sub> emissions. A data-driven model always provides more accurate results for the given range of observations and has unknown extrapolation properties, while a physical model exhibits better generalization capability. According to the model characteristics, the physical model is category-specific, and the data-driven model is vehicle-specific. When the historical driving data of the vehicle of interest are sufficient and meet the requirements of the training set, the data-driven model is more appropriate for estimating the real-time CO<sub>2</sub> emissions. Otherwise, the physical model is preferable. The detailed modelling process and model structures are introduced in the following section.

In an actual application, photos obtained from a roadside camera are sent to an image processing system, which is capable of identifying the make and model of the vehicle to be estimated. Then, the parallel system selects the suitable model type. The system adopts the physical model of the corresponding category (light-/heavy-duty gasoline or light-/heavy-duty diesel) with the default parameters (sometimes a quick calibration step is needed to improve model accuracy) when the historical dataset is insufficient and some basic vehicle parameters are missing. When the CO<sub>2</sub> emissions of each vehicle have been estimated for a certain investigation region, the overall carbon emissions can be obtained. Combined with the gas emission dispersion model, the total CO<sub>2</sub> emission rate is compared with the measurement of the roadside CO<sub>2</sub> smart sensors. If the estimated deviation



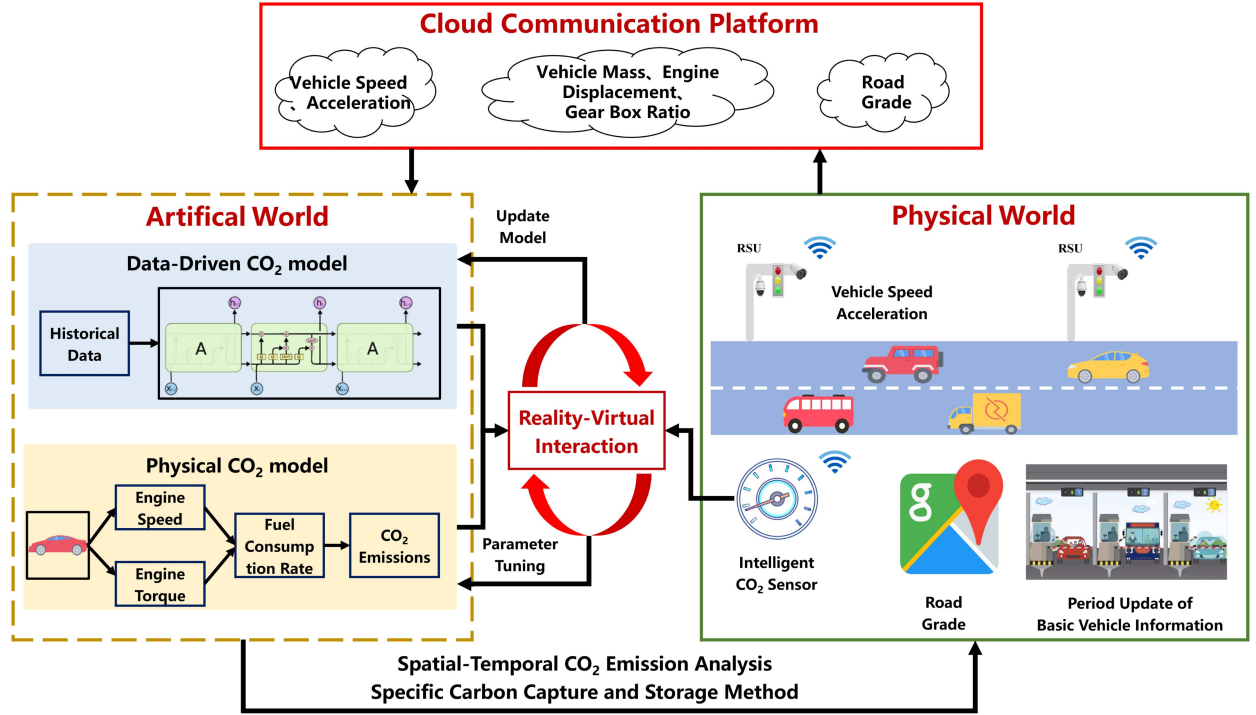


Fig. 2. The technical framework of the proposed parallel supervising system.

exceeds the threshold value, the model is updated online through reality-virtual interaction. Specifically, the physical model tunes its primary parameters to improve its estimation accuracy, while the data-driven model executes continued training from the base model. Based on this self-evolutionary mechanism, this parallel supervision system is capable of updating the model online and accumulating useful datasets. Unlike most previous studies, this paper focuses on second-by-second CO<sub>2</sub> emissions rather than average CO<sub>2</sub> emissions (by time or distance), which are more valuable for further spatial-temporal analyses.

### B. Combined Model Structure

This subsection introduces the theoretical background of the combined model structure. In this study, we focus on the relationships between CO<sub>2</sub> emissions and simple kinematic variables such as instantaneous speed, acceleration, and road gradients. As will be discussed later, the physical CO<sub>2</sub> model adopts a cascaded modeling methodology, while the data-driven model is based on an LSTM neural network.

1) **Physical CO<sub>2</sub> Model:** To achieve improved estimation accuracy, the combustion process that occurred in the engines should be simulated to capture the CO<sub>2</sub> generation mechanism. The physical CO<sub>2</sub> model adopts a cascaded structure, which selects engine speed and engine torque as intermediate (proxy) variables to accurately capture vehicle dynamics, as shown in Fig. 3. Traditional power-based methods exhibit some deficiencies in model accuracy, and engine-level variables may be employed as complementary features [21] and [31]. As engine speed and torque are difficult to measure without specialist equipment in real-world situations, two submodels are developed to estimate these two variables online. Specifically,

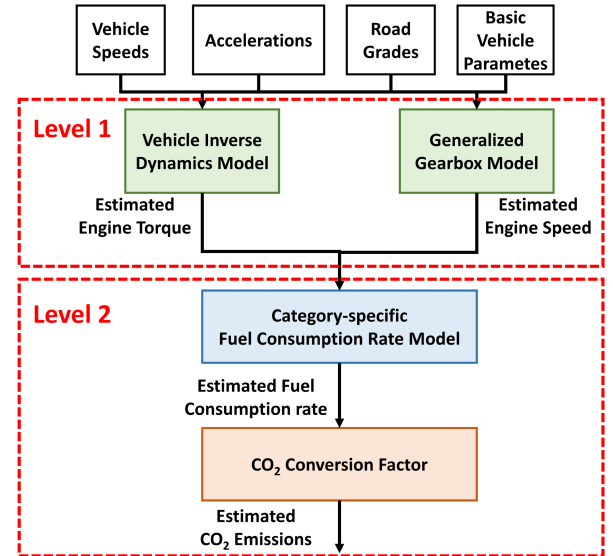


Fig. 3. Flowchart representation of cascaded modeling methodology.

an inverse vehicle dynamics model is used to estimate engine torque, while a generalized gearbox model is adopted to estimate engine speed. Thus, a cascaded structure is made, containing an upper level to estimate engine-related variables and a lower level to estimate real-time vehicle CO<sub>2</sub> emissions. The primary information required by the physical model is listed in Table I. As the physical model is based on the operating mechanisms of vehicles and engines, they can be considered category-specific (light-/heavy-duty gasoline or light-/heavy-duty diesel) models. To further improve the model accuracy, simple calibration is needed to fine-tune the critical parameters (such as gear ratios).

TABLE I  
PRIMARY INFORMATION REQUIRED BY THE PHYSICAL MODEL (FCRM  
DENOTES FUEL CONSUMPTION RATE MAP)

| Information                  | Form      | Information Source         |
|------------------------------|-----------|----------------------------|
| Speed                        | Input     | GPS                        |
| Acceleration                 | Input     | GPS                        |
| Road gradient                | Input     | HD map                     |
| Vehicle mass                 | Parameter | Weighting at toll stations |
| Frontal area                 | Parameter | Certification Manual       |
| Wheel effective radius       | Parameter | Certification Manual       |
| Gear ratios                  | Parameter | Certification Manual       |
| Engine size                  | Parameter | Certification Manual       |
| Rolling friction coefficient | Parameter | Certification Manual       |
| FCRM                         | Parameter | Category-specific FCRM     |

TABLE II  
PRIMARY INFORMATION REQUIRED BY THE DATA-DRIVEN MODEL

| Information   | Form  | Information Source |
|---------------|-------|--------------------|
| Speed         | Input | GPS                |
| Acceleration  | Input | GPS                |
| Road gradient | Input | HD map             |

The primary information required by the data-driven model is listed in Table II. It should be noted that the model inputs do not include the vehicle mass (which has a significant impact on carbon emissions, especially for heavy-duty vehicles), mainly because insufficient driving data with different loads are available. Therefore, the obtained data-driven model is vehicle-specific and is only valid for full-load conditions. To address this issue, real-world road tests are in progress to enrich the vehicle driving data under different loads (from no load to full load). Furthermore, according to the combined model structure, this supervision system can activate the physical model when the vehicle mass is obviously different from the training dataset (similar to this test dataset). Therefore, the physical model is so-called category-specific, while the data-driven model is vehicle-specific.

The hidden state  $c$  is controlled by a structure called a “gate,” consisting of a forget gate, an input gate, and an output gate. The forget gate mainly selectively forgets the output of the previous moment  $h_{t-1}$  and only retains the key historical information [35]. It controls the historical information retention strength of the cell’s hidden state. The input gate mainly selectively memorizes the input at the current moment  $X_t$ , which controls the intensity of the hidden state input. The output gate considers all the calculated results and generates an output at the current moment  $h_t$  for the LSTM cell. The calculation formula for the LSTM cell is as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, X_t] + b_f), \quad (1)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, X_t] + b_i), \quad (2)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, X_t] + b_c), \quad (3)$$

$$c_t = f_t \times c_{t-1} + i_t \times \tilde{c}_t, \quad (4)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, X_t] + b_o), \quad (5)$$

$$h_t = o_t \times \tanh(c_t), \quad (6)$$

where  $W$  is the weight matrix and  $b$  is the bias term.

#### IV. DETAILED MODELING APPROACHES

As the combined CO<sub>2</sub> estimation model is the most important component of the whole supervision system, this section describes the process of establishing the physical and data models in detail.

##### A. Details of the Physical CO<sub>2</sub> Model

The physical CO<sub>2</sub> model consists of an inverse vehicle dynamics model, a generalized gearbox model, a generalized fuel consumption map, and a CO<sub>2</sub> conversion factor, which are introduced in a step-by-step manner in this subsection. First, the inverse vehicle dynamics model is based on the longitudinal

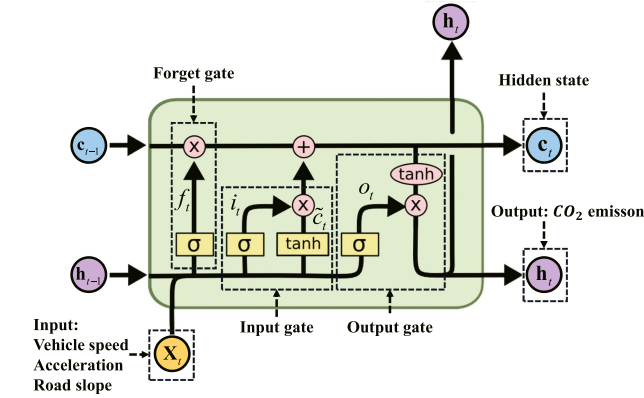


Fig. 4. Schematic block diagram of LSTM for predicting vehicle CO<sub>2</sub> emissions.

The inverse vehicle dynamics model is based on vehicle resistance and acceleration requirements. Air drag and friction resistance can be calculated using the drag and rolling coefficients, while gradient resistance can be calculated based on the road gradients [32]. In general, the selected gear shifting strategy depends on vehicle speeds and accelerator pedal positions, and this always requires a vehicle-specific transmission map [33] and [34]. To develop a simple generalized gearbox model, a one-parameter shifting strategy (only related to vehicle speeds) is adopted in this paper, and the gear shifting strategy is determined by the selected vehicle and transmission system categories.

On this basis, a generalized FCR map and a CO<sub>2</sub> conversion equation are built in level 2 to estimate real-time CO<sub>2</sub>. Due to the large variations in the FCR characteristics of different fuel types, a category-specific (light-/heavy-duty gasoline or light-/heavy-duty diesel) FCR model (FCRM) is designed to estimate the real-time FCR. Finally, a typical CO<sub>2</sub> conversion factor is adopted to estimate the CO<sub>2</sub> emission rate from the FCR.

2) *Data-Driven CO<sub>2</sub> Model*: The proposed data-driven CO<sub>2</sub> model in this paper is based on an LSTM neural network. The LSTM model improves upon the RNN structure and is often used to fit and predict time series data. Compared to a traditional RNN, the LSTM model can transmit more sophisticated information when dealing with time series data. The structure of the LSTM model is shown in Fig. 4, where  $X$  represents the model input,  $h$  represents the model output,  $c$  is the hidden state of the LSTM cell,  $\sigma$  represents a fully connected network, and  $\times$  is the multiplication operation.

force of the vehicle. The air drag resistance, friction resistance, road slope resistance, and acceleration requirements are considered in this module [36]. To reflect the internal friction resistance of the vehicle, an engine speed-related term is added. The total driving force is calculated as:

$$F_{total}(k) = \frac{1}{2}C_D A \rho v^2(k) + M_v g (f \cos(\alpha(k)) + \sin(\alpha(k))) + M_v a(k) + C_{IF} ES(k) \quad (7)$$

where  $C_D$  is the aerodynamic drag coefficient,  $A$  is the frontal area,  $\rho$  is the density of the ambient air,  $M_v$  is the vehicle mass,  $f$  is the rolling friction coefficient related to the tire and road condition,  $g$  is the acceleration of gravity,  $\alpha(k)$  is the slope angle in radians,  $a$  is the instantaneous acceleration,  $C_{IF}$  denotes the internal friction coefficient related to the engine speed, and  $ES$  is the engine speed. Time-related variables are denoted by  $(k)$ , and they are obtained online through cloud communications, while the other variables are fundamental vehicle-specific parameters.

Engine torque is the source of the driving force, which is transmitted through a complex drive train. Thus, the engine torque ( $T_e$ ) can be estimated by the equation (8).

$$T_e(k) = \frac{F_{total}(k)r_w}{\eta_t I_o I_g(k)} \quad (8)$$

where  $\eta_t$  is the mechanical efficiency of the driveline,  $I_o$  is the transmission ratio of the final gears,  $I_g$  is the transmission ratio of the current gear, and  $r_w$  is the effective wheel radius.

As the detailed shifting logic is hard to acquire, the generalized gearbox model utilizes a one-parameter shifting strategy (only related to speed). Different vehicle speed regions correspond to different gear ratios, while the specific gear ratios of different gears can be obtained through the vehicle certification manual. Thus, the engine speed  $\omega(k)$  can be estimated by equation (9).

$$\omega(k) = \frac{30v I_o I_g}{\pi r_w} \quad (9)$$

According to the cascaded architecture, the second level estimates the FCR from the estimated engine speed and engine torque (from level 1). The category-specific FCRM adopts a simple yet accurate model structure, which is category-specific and modified by an engine size-related coefficient. The category-specific FCRM is fitted by a two-dimensional polynomial, as in [37]:

$$\dot{m}_{fuel} = \sum_{i=0}^2 \sum_{j=0}^2 \kappa_{i,j}(ES) T_e^i(k) \omega^j(k) \quad (10)$$

where  $\kappa_{i,j}(ES)$  denotes the fitting coefficients related to engine size ( $ES$ ), and  $\dot{m}_{fuel}$  is the FCR with units of [g/s]. It is worth noting that a set of  $\kappa_{i,j}(ES)$  is only valid for an engine with a certain fuel type. Therefore, this model should be switched to the corresponding version in real applications.

CO<sub>2</sub> emissions are linearly related to the FCR, and the CO<sub>2</sub> conversion factor is determined by the engine type (gasoline or diesel), as follows:

$$\dot{m}_{CO_2} = T_{fuel} \dot{m}_{fuel} \quad (11)$$

TABLE III  
PRIMARY PARAMETERS OF TWO TESTING VEHICLES

|                                | LDGV  | HDDV   |
|--------------------------------|-------|--------|
| Vehicle mass (kg)              | 1470  | 46500  |
| Engine size (L)                | 2.0   | 13.0   |
| Rate power (kw)                | 108   | 418    |
| Frontal area (m <sup>2</sup> ) | 1.7   | 4.2    |
| Wheel radius (m)               | 0.364 | 0.572  |
| Rolling friction coefficient   | 0.011 | 0.0074 |
| Aerodynamic drag coefficient   | 0.36  | 0.5    |

where  $\dot{m}_{CO_2}$  is the CO<sub>2</sub> emission rate with units of [g/s] and  $T_{fuel}$  is a fuel-related conversion factor;  $T_{gasoline}$  is 3.15 and  $T_{diesel}$  is 3.06 in China [38].

### B. Details of the Data-Driven CO<sub>2</sub> Model

The data-driven CO<sub>2</sub> model is established based on the LSTM model with vehicle speed, acceleration, and road gradient information as inputs and vehicle CO<sub>2</sub> emissions as outputs. Taking the impact of the vehicle's historical state on CO<sub>2</sub> emissions into account, the LSTM model uses historical data sequences over a given horizon  $n_p$  as the training inputs [39]. The training inputs and training output of the LSTM model at time  $k$  can be expressed as:

$$\mathbf{X}_k = \begin{bmatrix} v(k - n_p + 1) & v(k - n_p + 2) & \cdots & v(k) \\ a(k - n_p + 1) & a(k - n_p + 2) & \cdots & a(k) \\ \alpha(k - n_p + 1) & \alpha(k - n_p + 2) & \cdots & \alpha(k) \end{bmatrix}, \quad (12)$$

$$\mathbf{h}_k = \dot{m}_{CO_2}(k). \quad (13)$$

To avoid overfitting the LSTM model during training, multiple pieces of real data with different lengths are selected as the training set based on the random sampling method. Then, the Adam optimizer is adopted to minimize the training errors, and the data-driven CO<sub>2</sub> model can be acquired after reaching the maximum number of iterations.

## V. RESULTS AND DISCUSSION

### A. Real-World Tests on an LDGV and an HDDV

To validate the proposed parallel supervision system, real-world tests are conducted on a light-duty gasoline vehicle (LDGV) and a heavy-duty diesel vehicle (HDDV). To reflect the representative working conditions of both vehicles, the test LDGV travels over urban overpasses, while the test HDDV operates on highways. The primary parameters of the two testing vehicles are listed in Table III. Some real-time measurement tools, including a fuel consumption meter, a GPS, an INS, a Mobileye camera, human-machine interface, and a vehicle gateway, are installed on the test LDGV (Fig. 5) [40]. The experimental test data are collected for the subsequent offline validation process with a sampling interval of 1 s.

Vehicle speeds, accelerations, and road gradients are adopted as the model inputs, while FCRs are considered the experimental findings. As vehicle CO<sub>2</sub> emissions significantly relate to the corresponding FCR (Section 4.B), no additional specialized equipment is used to directly measure CO<sub>2</sub> emission rates, and

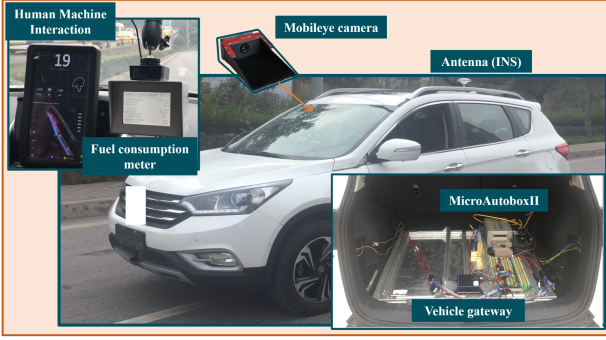


Fig. 5. The test LDGV with real-time measurement devices.

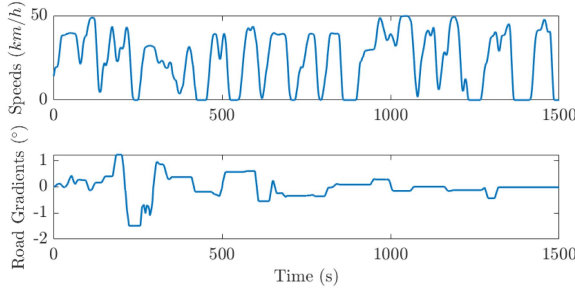


Fig. 6. The specific driving cycle of the LDGV.

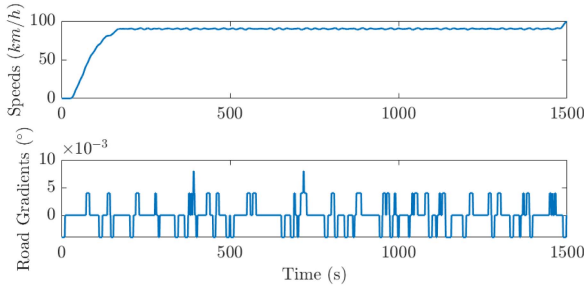


Fig. 7. The specific driving cycle of the HDDV.

the “Measured CO<sub>2</sub> emissions” term used in this section denotes the product of the measured FCR and the conversion factor. The detailed driving cycles (time histories of the speed, acceleration, and road gradients) of the LDGV and HDDV tests are shown in Fig. 6 and Fig. 7, respectively. To quantitatively compare the estimated results obtained for the two sets, the lengths of both tests are set to be 1500 seconds.

To evaluate the model accuracy, the coefficient of determination ( $R^2$ ) and mean squared error (MSE) are used as comparison indicators. These values are calculated using equations (14) and (15), respectively:

$$R^2 = 1 - \frac{\sum_{n=1}^{N_i} (x_i - \hat{x}_i)^2}{\sum_{n=1}^{N_i} (x_i - \bar{x})^2} \quad (14)$$

$$MSE = \frac{\sum_{n=1}^{N_i} (x_i - \hat{x}_i)^2}{N_i} \quad (15)$$

where  $x$  denotes the variables to be estimated (engine speed, engine torque, and CO<sub>2</sub> emission rate),  $x_i$  and  $\hat{x}_i$  are the  $i$ th real and estimated values, respectively,  $N_i$  is the total number of

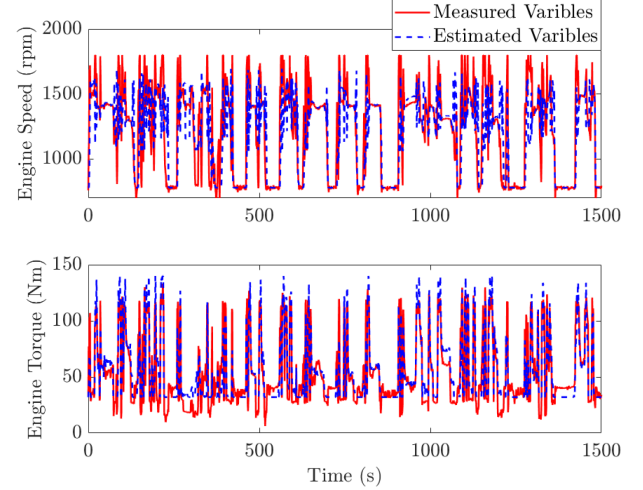
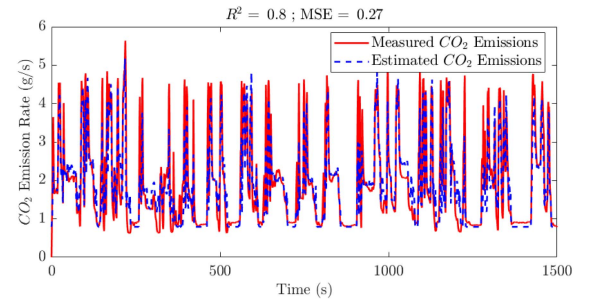


Fig. 8. The estimation of engine speeds and engine torques under the LDGV data set.

Fig. 9. The estimation of CO<sub>2</sub> emissions based on the physical model under the LDGV data set.

sampling points in each testing cycle, and  $\bar{x}$  is the average value of the associated variable.

### B. Results of the Physical Model

1) *CO<sub>2</sub> Estimation for the LDGV*: The cascaded physical model selects engine speed and torque as the proxy variables, which are the outputs of the level-1 layer and the inputs of the level-2 layer. To fully demonstrate the performance of the model and locate the causes of errors, the estimations of the engine speed and torque are compared first (Fig. 8). Overall, the accuracy of the engine state estimations is sufficient for meeting the requirements of subsequent CO<sub>2</sub> estimations, but some deficiencies are also present in the high-speed and low-torque regions. Estimating the engine statuses in these regions is very hard, mainly due to the high dynamics within these operating points. Although the specific FCR (g/Kwh) is high at these operating points, the required power (Kw) and the percentage of the total driving cycle are small, so we can disregard the impact of this part [37]. In addition, the simplified one-parameter shifting strategy leads to inaccurate transmission ratio estimation to some extent, which eventually results in underestimated engine speed and overestimated engine torque.

Based on the estimated engine speed and engine torque, the CO<sub>2</sub> emissions can be obtained from the second level, and the results are shown in Fig. 9. To realize comparisons between



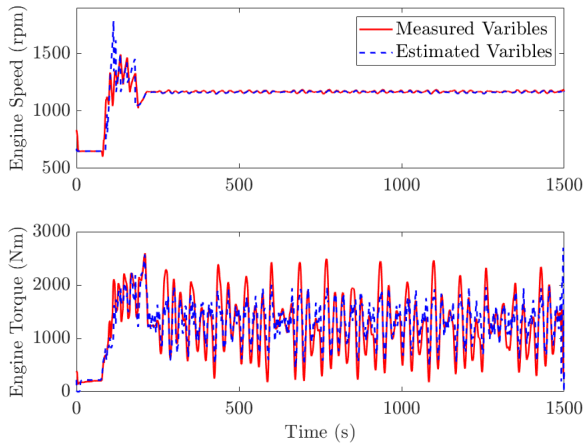


Fig. 10. The estimation of engine speeds and engine torques under the HDDV data set.

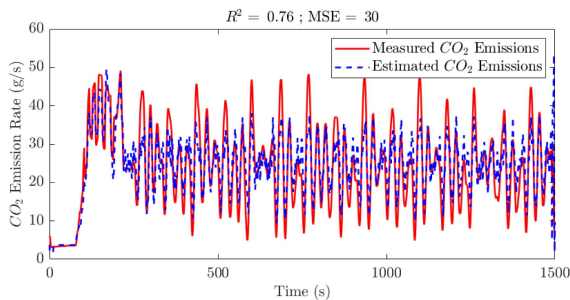


Fig. 11. The estimation of CO<sub>2</sub> emissions based on the physical model under the HDDV data set.

different methods, the correlation coefficient ( $R^2$ ) and MSE are adopted as model accuracy evaluation metrics. In this scenario, the  $R^2$  is 0.80, and the MSE is 0.27, validating the effectiveness of the physical model. As all model inputs are easy to obtain, this proposed physical model can be applied at a large scale with satisfactory precision.

2) *CO<sub>2</sub> Estimation for the HDDV*: To further validate the universality of the physical model, another real-world test is conducted with an HDDV, which has a different engine type with a drastically different engine size. Furthermore, some primary parameters are unknown for this HDDV, especially the transmission system-related parameters such as the gear ratios and shifting speeds. The category-specific (HDDV) parameters are adopted as their default values, and a simple calibration process is carried out based on the testing data.

The estimated engine speed and engine torque results are demonstrated in Fig. 10. The overall performance is close to that in gasoline engine results, while relatively high deficiencies occur in the gear shifting process. This is mainly due to the limited prior knowledge about the transmission system. The adopted transmission parameters are modified from their category-specific default values, and this step may introduce some uncertainties. In relatively steady regions, the estimated engine variables are much more accurate.

In Fig. 11, the estimated CO<sub>2</sub> emissions of this heavy-duty vehicle are compared with the experimental results. Specifically, the  $R^2$  is 0.76, and the MSE is 30, and the accuracy is acceptable

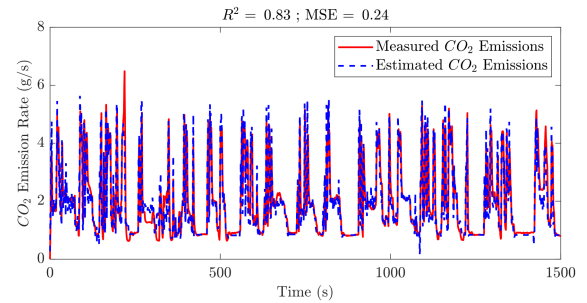


Fig. 12. The estimation of CO<sub>2</sub> emissions based on the data-driven model under the LDGV data set.

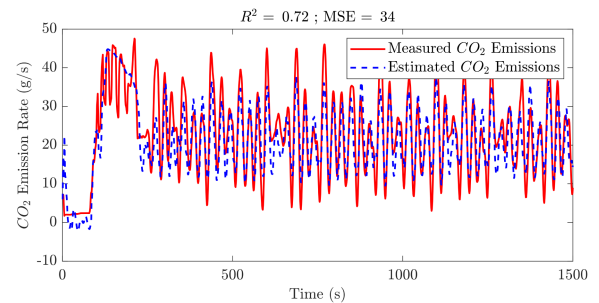


Fig. 13. The estimation of CO<sub>2</sub> emissions based on the data-driven model under the HDDV data set.

given the limited information. The sufficient accuracy obtained on this HDDV dataset is important since it further verifies the robustness of the physical model. In other words, the physical model is capable of estimating the CO<sub>2</sub> emissions of unfamiliar vehicles even with some unknown primary parameters. This is critical for enabling the supervision system to adapt to the modern transportation environment, as it includes various vehicle categories with different technologies.

### C. Results of the Data-Driven Model

For valid comparisons with the physical model, the CO<sub>2</sub> estimation ability of the data-driven model is analyzed. To represent the regular driving conditions, other 1500 s dataset (also from LDGV and HDDV) are used for model training, respectively.

1) *CO<sub>2</sub> Estimation for the LDGV*: In Fig. 12, the CO<sub>2</sub> emissions of gasoline vehicles estimated based on the data-driven model are compared with the experimental results. In this scenario, the  $R^2$  is 0.83, and the MSE is 0.24, making this model slightly more accurate than the physical model. It should be noted that some small fluctuations are observed in the estimated results. This is mostly due to the noise in the traffic data having a significant impact on the training and prediction processes of the LSTM model.

2) *CO<sub>2</sub> Estimation for the HDDV*: To further validate the effectiveness of the data-driven model, the HDDV dataset is also used for an analysis based on the data-driven model. The related results are shown in Fig. 13; the  $R^2$  is 0.72, while the MSE is 34. Compared with the physical model, the data-driven is not only has lower evaluation indices but also is significantly weak in terms of capturing the transient characteristics (as shown in the first 100 seconds of the experiment). The HDDV test operates



TABLE IV  
OVERALL COMPARISONS OF THE COMBINED MODEL UNDER LDGV AND HDDV DATA SETS

|      | Physical model |      | Data-driven model |      |
|------|----------------|------|-------------------|------|
|      | $R^2$          | MSE  | $R^2$             | MSE  |
| LDGV | 0.8            | 0.27 | 0.83              | 0.24 |
| HDDV | 0.76           | 30   | 0.72              | 34   |

primarily on highways, and the vehicle operates mostly at a constant cruising speed, resulting in a lack of data diversity and relatively few transient processes, as shown in Fig. 7. These deficiencies lead to some shortcomings in the training dataset, which causes a decrease in the estimation accuracy. The limited diversity of the historical data obviously deteriorates the model performance. Therefore, the parallel supervision system prefers to select physical models to estimate real-time vehicle CO<sub>2</sub> emissions when the given historical data are insufficient, even if some required HDDV parameters (gear ratios) are unknown.

#### D. Overall Performance of the Combined CO<sub>2</sub> Emission Model

Within the scope of this paper, both the physical and data-driven models provide accurate estimations of vehicle CO<sub>2</sub> emissions, as shown in Table IV. When the historical data are sufficient, the data-driven model yields more accurate estimation results than the physical model (under the LDGV dataset). However, when the historical data are insufficient in size and diversity, the data-driven model exhibits some deficiencies, while the physical model presents more robustness even when some primary parameters are unknown (under the HDDV dataset). As the data-driven model is vehicle-specific but the physical model is category-specific, the physical model is capable of acting as a default model that guarantee the output estimations for unfamiliar vehicles. Moreover, in the discussion section of this paper, the modeled and validated vehicles are identical, which is suitable for a data-driven model. If the comparison is extended to a category-level scenario (or different vehicles), the physical model may exhibit more obvious advantages. In summary, the combined CO<sub>2</sub> model is developed to adapt to different situations (regarding the availability of sufficient historical data), and it can leverage both its learning ability and physics-based knowledge.

#### E. Deterioration Factor of Vehicle CO<sub>2</sub> Emissions

Engine-out NO<sub>x</sub>, HC, CO, and PM deterioration factors (DFs) have been investigated by some authors, and these factors have also been considered in emission regulations [41] and [42]. To our knowledge, few studies have been conducted on the DF of vehicle CO<sub>2</sub> emissions, but this deterioration phenomenon will obviously affect the prediction accuracy of the model. An endurance test is conducted in an engine test bench with a modern GDI engine, and the experimental results are shown in Fig. 14. To simplify the test procedure, the engine speed is fixed at 2000 rpm, and the engine torque is set at its maximum value. During this near-400-hour accelerated aging test, the test engine exhibits power reduction to some extent, and the FCR shows an increasing tendency. Reduced turbocharger performance, engine

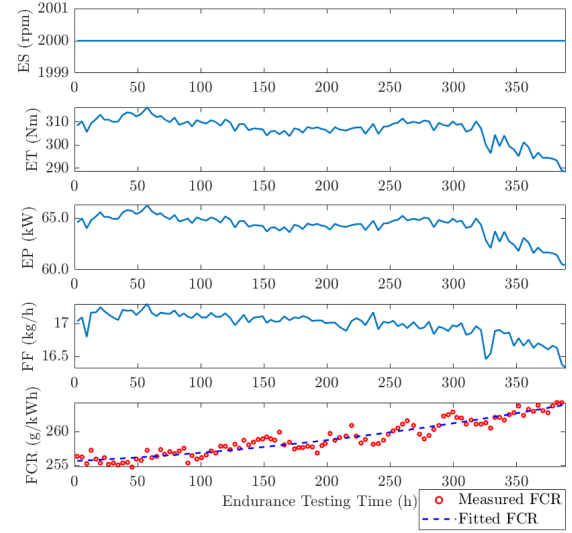


Fig. 14. Experimental results of the endurance test, ES denotes engine speed, ET denotes engine torque, EP denotes engine power, FF denotes fuel flow, FCR denotes fuel consumption rate.

friction, and carbon accumulation in the engine cylinders are the primary sources of the deteriorated fuel efficiency.

According to previous experience, the DF of vehicle CO<sub>2</sub> emissions always stabilizes after a 400-hour endurance test. Therefore, the DF is defined as a segmentation function with respect to the operation time, which is defined as follows:

$$DF = P_1 * t^2 + P_2 * t + P_3 \quad t \leq t_{critical}$$

$$DF = DF_{max} \quad t > t_{critical} \quad (16)$$

where DF is the multiplicative deterioration factor of vehicle CO<sub>2</sub> emissions,  $t$  is the equivalent engine operating time (transformed from the actual vehicle operating time) with units of [h],  $t_{critical}$  is the critical time point (400 full-load operations in this scenario),  $P_x$  denotes the fitting coefficients, and  $DF_{max}$  is the maximum DF with respect to the test engine (1.04 in this scenario). A polynomial function is adopted to match the experimental findings; the  $R^2$  of the fitted quadratic polynomial is 0.891, while the linear polynomial has an  $R^2$  of 0.877. The bottom subgraph of Fig. 14 shows the fitting comparisons, which can accurately reproduce the shapes and values. Vehicle CO<sub>2</sub> emissions are highly related to the FCR, so considering this DF in the estimation model will further improve the model accuracy.

## VI. CONCLUSION

To reduce CO<sub>2</sub> emissions in actual traffic scenarios, this paper establishes a parallel system to supervise vehicle CO<sub>2</sub> emissions on real roads with OBD-independent information. The whole system consists of three submodules in parallel, including the physical world, the artificial world, and a cloud communication platform. The physical world is the actual traffic environment with some necessary sensors, while the artificial world is mainly composed of the combined CO<sub>2</sub> estimation model. Reality-virtual interaction between the two worlds is capable of updating the model parameters online and accumulating useful data. The combined CO<sub>2</sub> model is the core part of the

whole system; it contains a physical module and a data-driven module. The supervision system chooses the appropriate model based on the availability of sufficient historical data. When the historical data are sufficient in size and diversity, the data-driven model is appropriate due to its more accurate estimation effect. Otherwise, the physical model is preferable because of its greater robustness. This novel combined structure is capable of adapting to various scenarios and realizing the supervision of the whole traffic system by using both its learning capacity and physics-based knowledge. The derivation processes for both models are introduced in this paper; the physical model adopts a cascaded structure, and the data-driven model is mainly based on an LSTM neural network.

To validate the model accuracy based on real-world experimental results, two real-world tests using an LDGV and an HDDV are conducted. When the given information is sufficient, both models achieve high accuracy, and the  $R^2$  values of the physical and data-driven models are 0.8 and 0.83, respectively (under the LDGV dataset). As the data diversity is limited in the HDDV dataset, and some primary parameters (gear ratios) are unknown, the estimation results show some deficiencies. The  $R^2$  values for the diesel vehicle are 0.76 (physical model) and 0.72 (data-driven model). Therefore, when the historical data are insufficient in size and diversity, the data-driven model exhibits some deficiencies, while the physical model presents more robustness even when some primary parameters are unknown. Moreover, as the data-driven model is vehicle-specific, whereas the physical model is category-specific, the physical model is capable of acting as a default model to guarantee the estimations obtained for unfamiliar vehicles (with limited historical data and some unknown parameters). In addition, a CO<sub>2</sub> emission DF model is developed based on the experimental results (accelerated aging test) to represent the performance of aged vehicles.

This paper focuses on proposing the framework of a parallel supervision system and the combined CO<sub>2</sub> estimation model. As this research is in progress, some shortcomings remain. The image processing system based on a roadside camera and an online updating mechanism (mentioned in the paper) will be investigated in the future. Moreover, this paper analyzes the main sources of vehicle CO<sub>2</sub> emissions (gasoline and diesel vehicles), but the real-world traffic environment is much more diverse, including natural gas vehicles, hybrid vehicles, electric vehicles, fuel-cell vehicles, etc. Thus, the vehicle category set should be enlarged, and the category-specific physical model should be complemented. The ability to reduce vehicle carbon emissions at multiple levels (traffic light adjustment, route planning, and driving habit advice) according to the obtained predictions is also an interesting research direction based on this supervision system. In addition, constructing this supervision system in a demonstration area (such as a campus or a community) will also be our research focus.

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smart automotive powertrains.

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